

Control-to-EU AI Act Mapping

Project 3: Governed Deployment Pattern (cross-cloud). Azure and AWS, mapped to Regulation (EU) 2024/1689.

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This matrix is the core of the governed deployment pattern. Each control traces to the specific EU AI Act obligation it satisfies and to the concrete Azure and AWS services that implement it, with an honest status for each. It rests on the current timeline: Article 50 transparency obligations are live from 2 August 2026, while the high-risk obligations (Articles 9 to 15) were deferred to 2 December 2027 by the Digital Omnibus. The pattern is built ahead of that deadline.

Control	EU AI Act obligation	Azure / AWS implementation	Status
Access and identity, least privilege	Art 15 (robustness and cybersecurity)	Azure: Entra managed identity, RBAC. AWS: scoped IAM execution roles	Built (P1 and P2)
EU data residency	Art 10 (data governance), with GDPR applying concurrently	Azure: Sweden Central, EU Data Boundary. AWS: Frankfurt, eu-central-1	Built (P1 and P2)
Content and safety filtering	Art 9 (risk management, mitigating harmful output)	Azure: AI Content Safety. AWS: Bedrock Guardrails content filters	Built (AWS, P2); specified (Azure)
PII detection and redaction	Art 10 (data governance), GDPR data minimisation	Azure: AI Language PII detection. AWS: Bedrock Guardrails PII masking, in and out	Built (AWS, P2); specified (Azure)
Human oversight, human-in-the-loop gate	Art 14 (human oversight)	Azure: approval step before action. AWS: manual approval task before a recommendation is acted on	Documented pattern
Record-keeping and audit logging	Art 12 (automatic logging over the lifecycle)	Azure: Azure Monitor, Log Analytics, diagnostic settings. AWS: CloudTrail, CloudWatch Logs, S3 record of inputs and outputs	Built in P3
Transparency	Art 50 (AI disclosure, live Aug 2026) and Art 13 (information to deployers)	Azure: article-level citations (P1). AWS: per-dimension justifications (P2), plus AI-use disclosure	Built (P1 and P2)
Accuracy, robustness, evaluation	Art 15 (accuracy, robustness) and Art 9 (testing)	Azure: Foundry evaluation, AI red teaming. AWS: ten-case eval harness, CloudWatch metrics, Guardrails Automated Reasoning checks	Built (P2 eval); specified (Azure)
Model routing and cost control	Operational governance (no direct Act obligation)	Azure: model tiering by use case. AWS: Bedrock Intelligent Prompt Routing	Demonstrated in P3
Risk classification and management	Art 6 (classification) and Art 9 (risk management system)	Cloud-agnostic process: classify risk tier, document risk-management steps	Documented process

Classification.

The two demo systems, the RAG advisor and the startup screener, classify as limited risk under the Act, in the transparency tier, not high risk, because neither falls into an Annex III category such as employment, credit scoring of individuals, or access to essential services. The pattern is nonetheless built to high-risk-grade controls (Articles 9 through 15), so it transfers directly to a high-risk deployment. Being able to say the systems are limited-risk while the pattern is engineered for high-risk, and to explain why each classifies as it does, is exactly the Article 6 reasoning a regulated employer is testing for.

Timing.

The transparency controls map to obligations already live from August 2026, while the human-oversight, logging, data-governance, and robustness controls map to the high-risk obligations landing December 2027. The pattern is deliberately built ahead of that deadline.

Precision.

Two rows are honestly qualified rather than overclaimed: EU residency is driven more by GDPR, which applies concurrently, than by a direct AI Act mandate, and model routing is operational governance with no specific

Act article. Flagging those rather than forcing every control onto an article is itself the credibility signal, because it shows where the law actually bites.